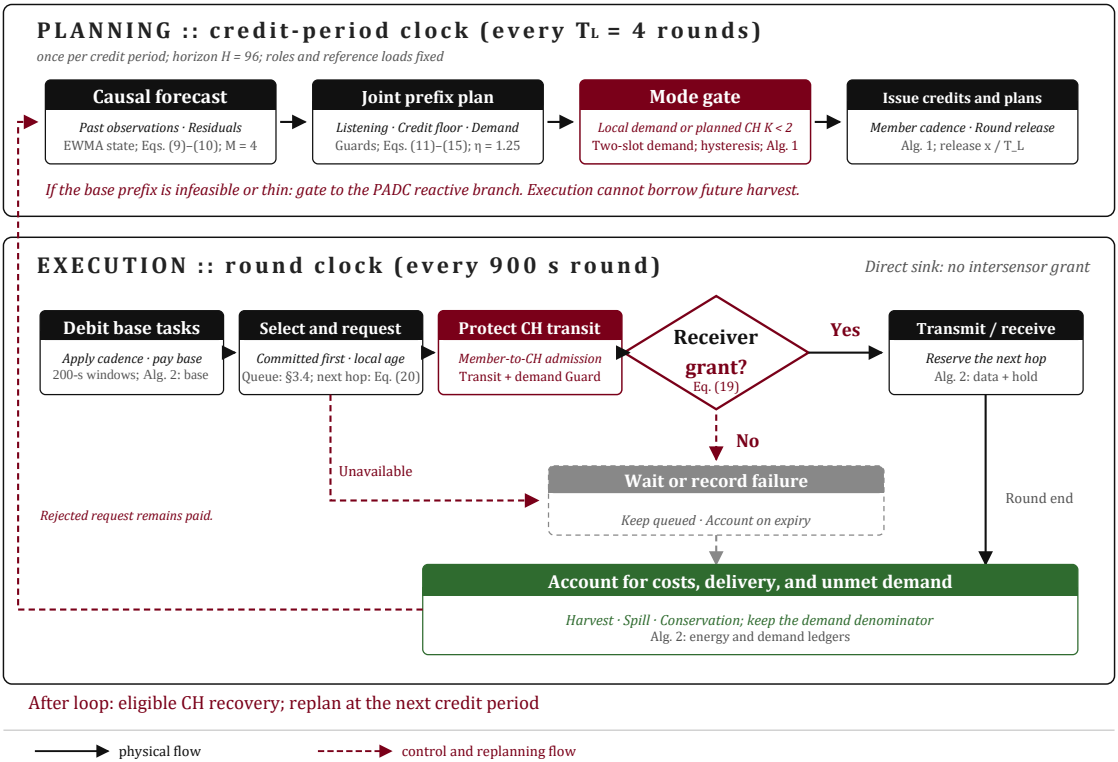

PLANNING :: credit-period clock (every T_L = 4 rounds)
once per credit period; horizon H = 96; roles and reference loads fixed
Causal forecast
Past observations · Residuals
EWMA state; Eqs. (9)–(10); M = 4
Joint prefix plan
Listening · Credit floor · Demand
Guards; Eqs. (11)–(15); η = 1.25
Mode gate
Local demand or planned CH K < 2
Two-slot demand; hysteresis; Alg. 1
Issue credits and plans
Member cadence · Round release
Alg. 1; release x / T_L
If the base prefix is infeasible or thin: gate to the PADC reactive branch. Execution cannot borrow future harvest.
EXECUTION :: round clock (every 900 s round)
Direct sink: no intersensor grant
Debit base tasks
Apply cadence · pay base
200-s windows; Alg. 2: base
Select and request
Committed first · local age
Queue: §3.4; next hop: Eq. (20)
Protect CH transit
Member-to-CH admission
Transit + demand Guard
Receiver grant?
Eq. (19)
Yes
Transmit / receive
Reserve the next hop
Alg. 2: data + hold
No
Unavailable
Wait or record failure
Keep queued · Account on expiry
Rejected request remains paid.
Round end
Account for costs, delivery, and unmet demand
Harvest · Spill · Conservation; keep the demand denominator
Alg. 2: energy and demand ledgers
After loop: eligible CH recovery; replan at the next credit period
physical flow
control and replanning flow